Git Workflow: Cloning, Managing Remotes & Pushing

This guide outlines how to clone an existing GitHub repository, make your own changes, and upload them to a **new, personal repository**. It also explains how to keep the connection to the original project as an `upstream` remote to easily fetch future updates.

1. Clone the Repository

Copy the clone URL of the original project from GitHub and download it to your local machine:

```
git clone <url-of-original-repo>
cd <folder-name>
```

2. Create a New Target Repository on GitHub

Create a new, completely empty repository on GitHub (do *not* include a README, `.gitignore`, or license). Copy the HTTPS or SSH URL of this new repository.

3. Configure Remotes (Two-Remote Setup)

By default, the `origin` remote points to the cloned original repository. To point your code to your own repository while still being able to fetch original project updates, we use two remotes:

1. **Define the original as `upstream`:**

   ```
   git remote rename origin upstream
   ```

2. **Add your own repository as the new `origin`:**

   ```
   git remote add origin <url-of-your-new-repo>
   ```

4. Modify Code and Push

Make your changes to the code, commit them locally, and upload them to your own repository:

```
git add .
git commit -m "Your commit message"
git push -u origin main
```

(Replace `main` if necessary with your branch name, e.g., `master`).

5. Pull Updates from the Original Repository (Optional)

If the original repository has been updated and you want to incorporate those changes into your code, you can synchronize them as follows:

```
git fetch upstream
git merge upstream/main
git push origin main
```